Western Mathematics Exams

2023
YEAR 11
EXAMINATION

Mathematics
Standard

SOLUTIONS

Multiple Choice Worked Solutions

No	Working	Answer
1	Absolute error $= \frac{1}{2} \times$ smallest unit $= 0.5$ seconds Measurement $=$ 1 minute and 20 seconds $= 80$ seconds Percentage error $= \frac{\text{Absolute error}}{\text{Measurement}} \times 100$ $= \frac{0.5}{80} \times 100$ $= 0.625\%$	**B**
2	If D is directly proportional to M, then $D = kM$, where k is a constant. The graph of this will be a straight line passing through the origin. Only option D meets both criteria.	**D**
3	**Theoretically** To find GST inclusive price $\times$ by 1.1 To Find answer after a 20% discount $\times$ by 0.8. So $X \times 1.1 \times 0.8 = X \times 0.8 \times 1.1$ since mult is commutative So they will get the same answer of $0.88X$. **Try a particular price of say \$60** Aimee adds GST then takes discount Price with $GST = 60 \times 1.1 = 66$ Price after taking 20% $discount = \$66 \times .8 = \$52.80$ Beatrix takes discount then adds GST Price after taking 20% $discount = \$60 \times .8 = \$48.00$ Price with $GST = 48.00 \times 1.1 = \$52.80$ They get the same price and would do so for any product and discount as the process is the same	**A**
4	All quarterly prices will be the same except the water usage. Water usage $= 45 \times 2.60 = \$117.00$ Quarterly cost $= 196.20 + 117.00 + 30.00$ $= \$343.20$ Annual cost $= 343.20 \times 4 = \$1372.8$ Weekly amount $= \$1372.80 \div 52 = \$26.40$	**D**
5	Group 1 | | Group 2 5 3 | 0 | 0 2 4 6 9 6 5 2 | 1 | 1 2 2 2 6 9 6 5 3 3 0 | 2 | 2 3 6 6 9 6 4 2 1 | 3 | 1 3 3 5 7 9 4 2 0 | 4 | 0 4 8 | 5 | 5 9 Number = 19 — Number = 25 (A is False) Median = 25 — Median = 26 (B is False) Mode = 23 — Mode = 12 (C is True) Range = 44 – 3 = 41 — Range = 59 – 0 = 59 (D is False)	**C**

6	For Group 2 Lower Q = $\frac{12+12}{2}$ = 12 Upper Q = $\frac{37+39}{2}$ = 38 Interquartile Range = 38 – 12 = 26	**B**
7	$\text{Total items} = 1.29 + 2.45 + 1.46 + 2.80 = 8.00$ $\text{P(Yuna)} = \frac{2.80}{8.00} = 0.35$ $\text{P(not Yuna)} = 1 - 0.35$ $= 0.65$	**C**
8	$\text{Diameter} = 50\text{ cm, so radius} = 25\text{ cm}$ $\text{Volume sphere} = \frac{4}{3}\pi r^3$ $= \frac{4}{3} \times \pi \times 25^3$ $= 65449.84694979$ $\text{Volume of hemisphere} = \frac{65449.84694979}{2}$ $= 32724.9234749\text{ cm}^3$ $= 32.7249\text{ litres}$ $= 32.7\text{ litres nearest 10th of a litre}$	**A**
9	$\text{Average weekly hours} = \frac{34+40+32+34}{4} = 35$ $\text{Weekly pay} = \frac{81900}{52} = 1575.00$ $\text{Hourly pay} = \frac{1575}{35} = \$45.00$	**B**
10	$m = \frac{\text{rise}}{\text{run}}$ $= -\frac{16}{4}$ $= -4$	**A**

2023 Year 11 Final Examination
Mathematics Standard

Name ______________________________ Teacher ______________________

Section I – Multiple Choice Answer Sheet

Allow about 25 minutes for this section
Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample:	2 + 4 =	(A) 2	(B) 6	(C) 8	(D) 9
		A ○	B ●	C ○	D ○

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ● B ● C ○ D ○

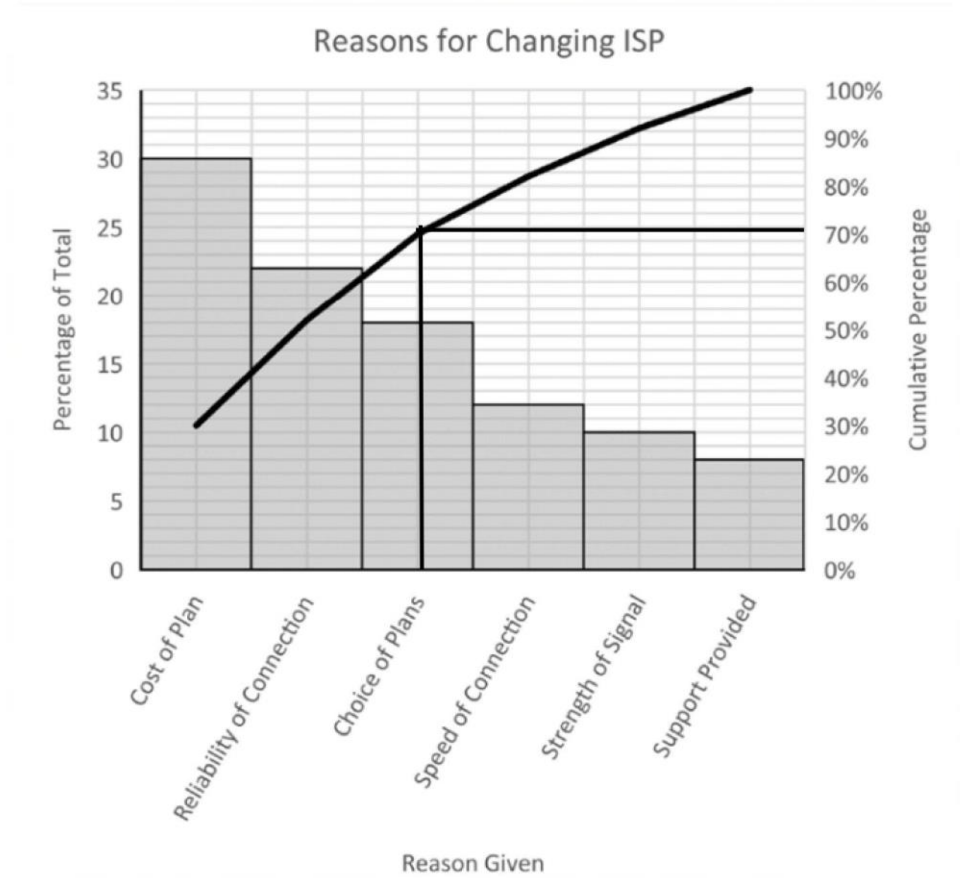
If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

correct

A ● B ● C ○ D ○

1.	A ○	B ●	C ○	D ○
2.	A ○	B ○	C ○	D ●
3.	A ●	B ○	C ○	D ○
4.	A ○	B ○	C ○	D ●
5.	A ○	B ○	C ●	D ○
6.	A ○	B ●	C ○	D ○
7.	A ○	B ○	C ●	D ○
8.	A ●	B ○	C ○	D ○
9.	A ○	B ●	C ○	D ○
10.	A ●	B ○	C ○	D ○

	Solution	Marks	Allocation of marks
Question 11			
(a)	$s = ut + \frac{at^2}{2}$ $s = 12 \times 4 + \frac{10 \times 4^2}{2}$ $= 128$	**1**	1 mark for correct answer
(b)	$s = ut + \frac{at^2}{2}$ $84 = 6u + \frac{3 \times 6^2}{2}$ $84 = 6u + 54$ $6u = 84 - 54$ $6u = 30$ $u = \frac{30}{6} = 5$	**2**	2 marks for correct answer 1 mark for correct substitution with error made in solution or equivalent merit.
Question 12			
	$\text{Time} = 4.35 \times 60 = 261$ seconds $\text{Distance} = 300000 \times 261 = 78\,300\,000$ $= 7.83 \times 10^7$	**2**	2 marks for correct answer is S N 1 mark if an error is made in calculation, or of not expressed correctly in S N.

<table>
<tr><td colspan="2">Question 13</td><td colspan="2"></td></tr>
<tr><td>(a)</td><td>

70% are addressed by covering the top 3 issues (Cost, Reliability and Choice)</td><td>1</td><td>1 mark for correct answer</td></tr>
<tr><td>(b)</td><td>Percentages for Technical issues = 22 + 12 + 10 = 44%</td><td>1</td><td>1 mark for correct answer</td></tr>
<tr><td colspan="4"></td></tr>
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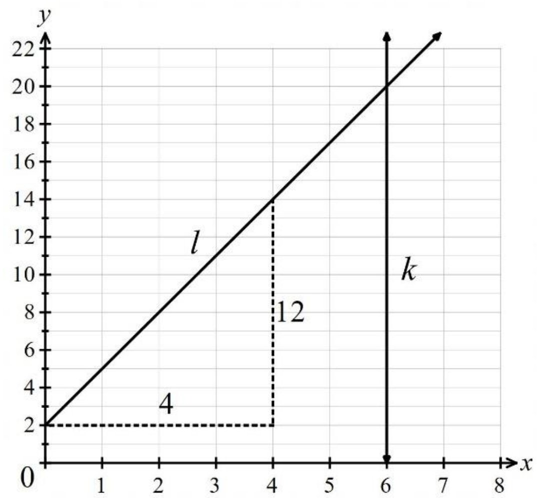
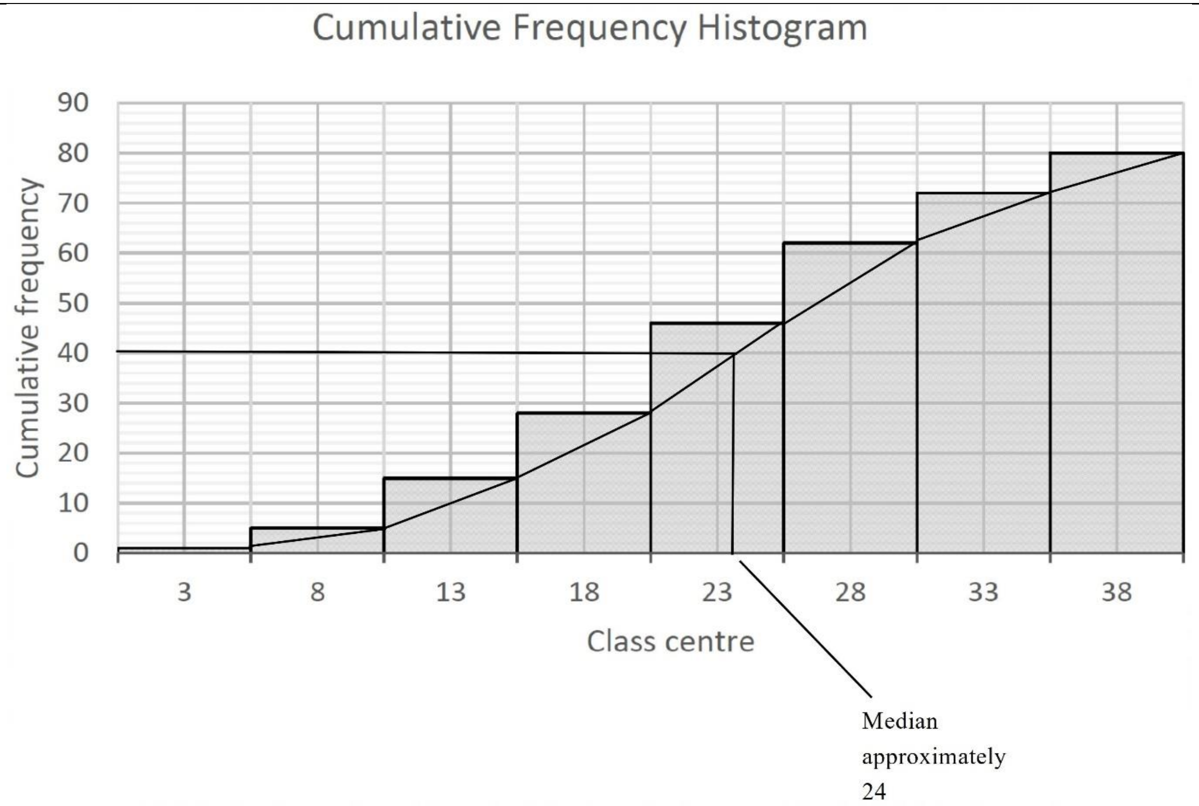
Question 14			
	$m = \dfrac{k^2 + wp}{d}$ $dm = k^2 + wp$ $dm - k^2 = wp$ $p = \dfrac{dm - k^2}{w}$	**2**	2 marks for correct answer 1 mark if a minor error is made in an otherwise correct worked solution or equivalent merit.

Question 15				
	Annual depreciation $= 12600 - 11592$ $= \$1008$ From Jan 2023 to Jan 2028 is 5 more years. $S = V_0 - Dn$ $S = 11592 - 5 \times 1008 = \$5040$ Value in Jan 2028 $= 11592 - 5040$ $= \$6552$	Annual depreciation $= 12600 - 11592$ $= \$1008$ Rate $= \dfrac{1008}{12600} = 0.08$ From 2022 to 2028 is 8 years Depreciation $(I) = PRN$ $= 12600 \times 0.08 \times 6 = 6048$ Value in Jan 2028 $= 12600 - 6048 = \$6552$	**2**	2 marks for correct answer 1 mark for correct annual depreciation or equivalent merit

Question 16			
	Area end $= 1.5 \times 1.6 + \dfrac{1}{2} \times 0.6 \times 1.6$ $= 2.88 \text{ m}^2$ Area sides (inc door) $= 2.5 \times 1.5 = 3.75 \text{ m}^2$ Area roof panel $= 2.5 \times 1.0 = 2.5 \text{ m}^2$ Total area $= 2 \times 2.88 + 2 \times 3.75 + 2 \times 2.5$ $= 18.26 \text{ m}^2$	**2**	2 marks for correct answer 1 mark for area calculation with minor error or missing panel or floor included or equivalent merit

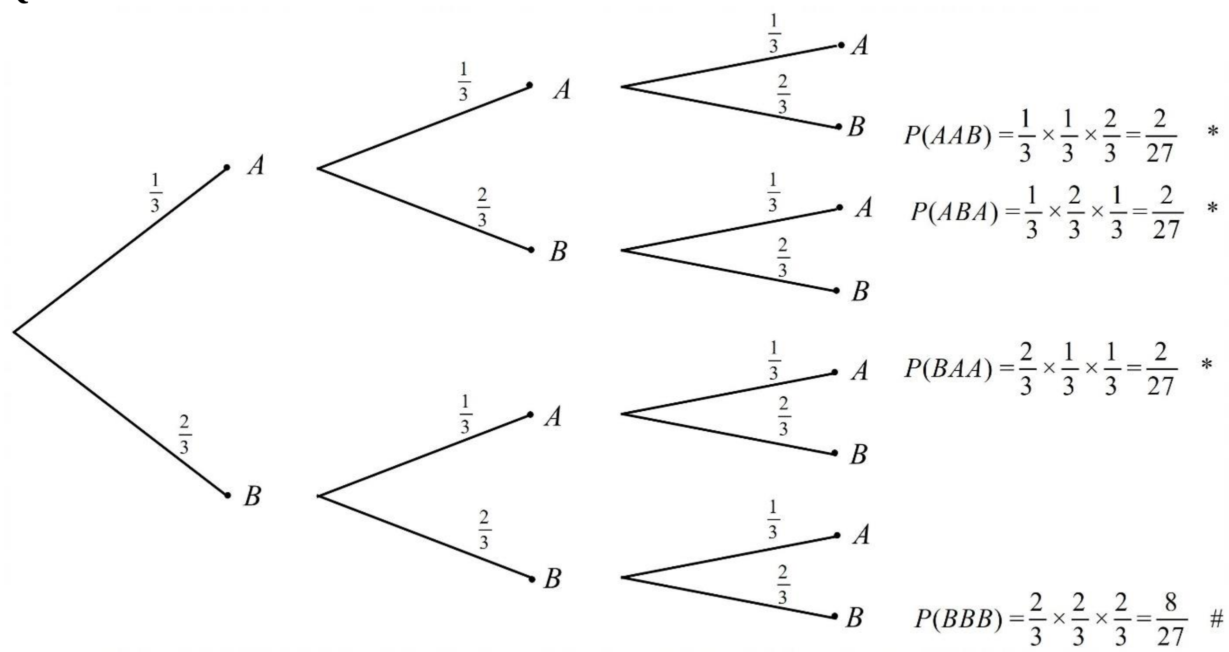
Question 17			
	John's results include five times as many trials as Greg's and so the relative frequency should be closer to the theoretical probability, because of the larger number of trials. The relative frequency using John's results for "3" is $\frac{68}{160} = \frac{17}{40} = 0.425$ If a student combines the results to increase the numbers, the relative frequency for "3" is $\frac{80}{192} = \frac{5}{12} = 0.417$ [Accept this as correct]	**2**	2 marks for correct explanation and probability 1 mark for correct explanation or correct probability only
Question 18			
(a)	Line k is a vertical line, which passes through 6 on the x-axis, so equation is $x = 6$.	**1**	1 mark for correct answer
(b)	Gradient of $l = \frac{\text{rise}}{\text{run}}$ $= \frac{12}{4}$ $m = 3$ Gradient can be found from any 2 points And using any form of the formula for gradient y intercept $= b = 2$ Gradient − intercept form of the equation is $y = mx + b$ Equation is $y = 3x + 2$ 	**2**	2 marks for correct equation 1 mark for correct gradient or equivalent merit

Question 19			
(a)	Number greater than 20 $= 18 + 16 + 10 + 8 = 52$ Percentage greater than 20 $= \frac{52}{80} \times 100 = 65\%$	**1**	1 mark for correct answer
(b)	$\Sigma fx = 1895$ Mean $= \frac{\Sigma fx}{\Sigma f} = \frac{1895}{80} = 23.6875$	**1**	1 mark for correct answer
(c)		**2**	2 marks for completed polygon and estimate from 23.5 to 24.0. I mark for minor error in polygon and median estimated correctly from polygon, or for correct polygon only

Question 20			
	Mass of one loaded weelbarrow $= 4.5 + 35 \times 3.8 = 137.5$ kg Crane limit $= 1.5$ tonnes $= 1500$ kg Number of loads $= \frac{1500}{137.5} = 10.90909$ (So, 11 would be over the load limit.) Number of wheelbarrows $= 10$ (which will fit on the platform as it holds up to 15)	**2**	2 marks for correct answer 1 mark for some correct and relevant calculation.
Question 21			
	$2000\ kj = 2000 \times 0.24$ Cal $= 480$ Cal Number of eggs $= 480 \div 78$ $= 6.15384615$ She should limit to 6 eggs	**2**	2 marks for correct answer 1 mark for correct number of Calories or equivalent merit
Question 22			
	Michelle Insurance $= 0.05 \times 46400 = \$2320$ Stamp Duty $= 1350 + 14 \times 5 = \$1420$ Total Cost $= 46400 + 2320 + 1420$ $= \$50140$ Simon Insurance $= 0.08 \times 44000 = \$3520$ Stamp Duty $= 440 \times 3 = \$1320$ Total Cost $= 44000 + 3520 + 1320$ $= \$48840$	**2**	2 marks for correct answer 1 mark for correct total answer for either person or for correct insurance for both, and minor error in stamp duty or equivalent merit

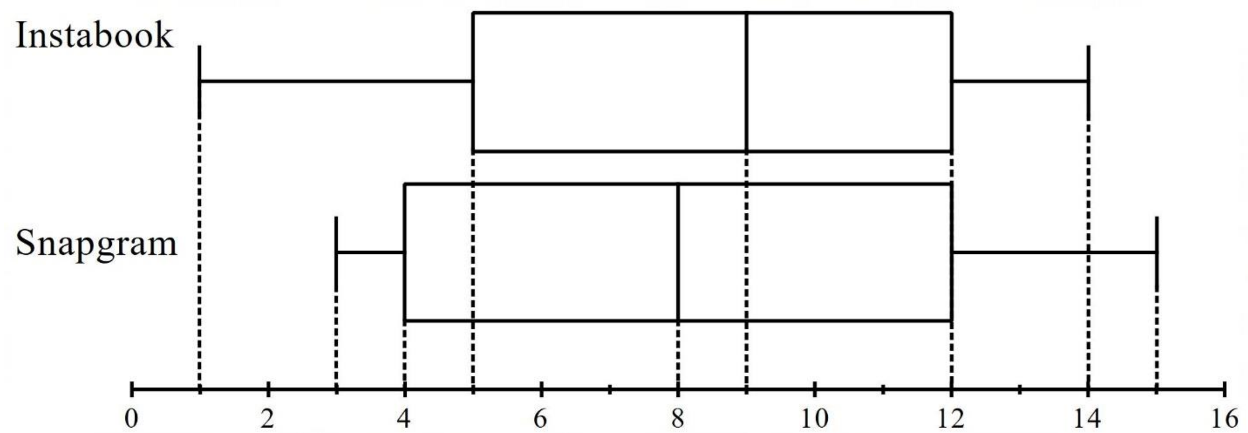
Question 23			
	$5g - 12 = 3(g + 6)$ $5g - 12 = 3g + 18$ $2g - 12 = 18$ $2g = 30$ $g = 15$	**3**	3 marks for correct answer 2 marks for solution with a minor error 1 mark for correct expansion or equivalent merit
Question 24			
	$\text{Total Hours} = 7 + 8 + 7 + 9 + 7 + 4 = 42 \text{ hours}$ $\text{Overtime hours} = 42 - 35 = 7 \text{ hours}$ $\text{Pay} = 35 \times 44.60 + 5 \times 44.60 \times 1.5 + 2 \times 44.60 \times 2$ $= \$\mathbf{2073.90}$ $\text{Or Equivalent hours} = 35 + 5 \times 1.5 + 2 \times 2 = 46.5 \text{ hours}$ $\text{Pay} = 46.5 \times 44.60 = \$\mathbf{2073.90}$	**2**	2 marks for correct answer 1 mark for correct total overtime hours and some relevant calculations or equivalent merit.

Question 25



(a)	Only one branch gives 3 B # $P(BBB) = \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{8}{27}$	**1**	1 mark for correct answer
(b)	Three branches have two A's and one B * Each has a probability of $\frac{2}{27}$ (see above) $P(2\ A's \text{ and one } B) = 3 \times \frac{2}{27} = \frac{2}{9}$	**2**	2 marks for correct answer 1 mark for correct probability of a relevant branch or equivalent merit

Question 26



(a)	Median(I) = 9 Median(S) = 8 Instabook is greater by 1 unit.	**1**	1 mark for correct answer
(b)	Range(I) = 14 – 1 = 13 Range(S) = 15 – 3 = 12 Instabook is greater by 1 unit.	**1**	1 mark for correct answer
(c)	Interquartile Range(I) = 12 – 5 = 7 Interquartile Range(S) = 12 – 4 = 8 Snapgram is greater by 1 unit.	**1**	1 mark for correct answer

Question 27				
(a)	Speed $= 90$ km/h $= 90\,000$ m/h $= \dfrac{90\,000}{60}$ m/min $= 1\,500$ m/min In 2 minutes it travels 3000 m	OR 2 minutes is $\frac{1}{30}$ of an hour Distance travelled $= \frac{90\,000}{30}$ m In 2 minutes it travels 3000 m	**1**	1 mark for correct answer
(b)	The car travels 3000 m in 2 minutes The car travels 3000 m in 120 seconds The car travels 100 m in 4 seconds (dividing by 30)		**1**	1 mark for correct answer
(c)	Speed $= 90$ km/h $D = 0.00001\, s^2$ $= 0.00001 \times 90^2$ $= 0.081$ km $= 0.081 \times 1000$ m $= 81$ m		**2**	2 marks for correct answer 1 mark for correct number of kilometres or equivalent merit

Question 28			
(a)	Monthly rate $= 4.8 \div 12 = 0.4\%$	**1**	1 mark for correct answer
(b)	$I = PRN$ For May I $= 12\ 193.15 \times 0.004 \times 1 = \$48.7726 = \$48.77$ (rounded) For June I $12\ 241.92 \times 0.004 \times 1 = \$48.96768 = \$48.97$ (rounded) May | \$12 193.15 | \$48.77 | \$12 241.92 June | \$12 241.92 | \$48.97 | \$12 290.89	**1**	1 mark for correct answer
(c)	Using simple interest $R = 0.048$ and $n = 0.5$ $I = 12000 \times 0.048 \times 0.5 = \$288.00$ Using compound Interest I $= 12290.89 - 12000$ $= \$290.89$ So earns an extra \$2.89	**1**	1 mark for correct answer

Question 29			
(a)	$Q = kT$ $140 = k \times 50$ $k = \frac{140}{50} = 2.8$ $Q = 2.8\ T$	**2**	2 marks for correct formula 1 mark for correct value of k or equivalent merit
(b)	In an hour Acme produces 240 litres (from graph or equation) In an hour Bingo produces $Q = 2.8 \times 60 = 168$ litres Difference in an hour $= 240 - 168 = 72$ litres.	**1**	1 mark for correct answer

Question 30			
(a)	Amount over threshold $= 50240 - 45000 = \$5240$ Income tax due $= 5092 + 0.325 \times 5240$ $= 5092 + 1703$ $= \$6\,795$	**1**	1 mark for correct answer
(b)	Medicare levy $= 50240 \times 0.02$ $= \$1\,004.80$ Total Tax Paid $= 310 \times 26 = \$8060$ Total Tax Due $= 6795 + 1004.80 = \$7799.80$ Tax refund $= \$8060 - 7799.80$ $= \$260.20$	**2**	2 marks for correct answer 1 mark for correct Medicare Levy only or for correctly calculated refund/bill found from incorrect levy or tax due
Question 31			
(a)	Plot 1 $A \approx \frac{40}{2}(90 + 124) + \frac{40}{2}(124 + 120)$ $\approx 4\,280 + 4\,880$ $\approx 9\,160 \text{ m}^2$	**1**	1 mark for correct answer
(b)	Plot 2 $A \approx \frac{50}{2}(120 + 140) + \frac{50}{2}(140 + 162)$ $\approx 6\,500 + 7\,550$ $\approx 14\,050 \text{ m}^2$ Proportion $= \frac{14\,050}{9\,160} = 1.53384279$ So one and a half times is more correct	**2**	2 marks for correct answer 1 mark for correct area only or for correct proportion found from incorrect area(s).

Question 32			
(a)	$P(\text{AB in 2019}) = 3.9 + 0.6 = 4.5\%$ $= 0.045$ $= \frac{9}{200}$	**1**	1 mark for correct answer
(b)	$P(+\text{ in 1994}) = 40.0 + 31.0 + 8.0 + 2.0 = 81.0\%$ $= 0.81 = \frac{81}{100}$ $P(+\text{ in 2019}) = 37.8 + 29.6 + 12.4 + 3.9 = 83.7\%$ $= 0.837 = \frac{837}{1000}$ A blood donor is 2.7% more likely to be positive in 2019, compared to 1994.	**2**	2 marks for two correct probabilities and a relevant comparison (difference not required) 1 mark for at least one correct and relevant probability and a comparison or equivalent merit.

Question 33			

Station	Arrival Times								
Charleston (Departure time)	3:00	3:30	4:00	4:30	5:00	5:30	6:00	6:30	7:00
Crystal Springs	3:25		4:25		5:25		6:25		7:25
Dingo Flat	3:39		4:39		5:39		6:39		7:39
Dry Lake	3:57		4:57		5:57		6:57		7:57
Arrowline	4:12	4:30	5:12	5:30	6:12	6:30	7:12	7:30	8:12
Boulder	4:42		5:42		6:42		7:42		8:42
Riverside	5:18		6:18		7:18		8:18		9:18
Bunkerville	5:30	5:42	6:30	6:42	7:30	7:42	8:30	8:42	9:30
Littlefield	5:54	6:04	6:54	7:04	7:54	8:04	8:54	9:04	9:54
Echo Ridge	6:21		7:21		8:21		9:21		10:21
St George	6:33	6:40	7:33	7:40	8:33	8:40	9:33	9:40	10:33

(a)	Need a time at St George before 8:10 which stops at Dry Lake, so must catch train at 4:57.	**1**	
(b)	To allow 10 minutes to meeting, must get to Arrowline Station at 5:35 at latest, so must catch the train that leaves Charleston at 4:30. Meeting ends at 6:15 and then a ten minute walk back to the station, so time is 6:25. The next train which stops at Echo Ridge leaves Arrowline at 7:12. (Arrives at 9:21)	**2**	2 marks for both correct times 1 mark for first correct time only or second 1 mark for correct answer correct time found from incorrect first time

Question 34			
(a)	Time difference = +10-(-3) = +13 hours so Sydney is 13 hours ahead of Brasilia. 13 hours before 5 pm Sunday 3rd is 4 am on Sunday 3rd	**1**	1 mark for correct answer
(b)	The plane arrives 18 and a half hours after 11:00 am on Tuesday 5th Sydney time so 5:30 am on Wednesday 6th Sydney time. 13 hours before 5:30 am on Wednesday 6th is 4:30 pm on Tuesday 5th Sept, Brasilia time.	**2**	2 marks for correct answer 1mark for correct arrival time in Sydney time or equivalent merit
(c)	$\text{Speed} = \text{dist/time}$ $= \dfrac{14220}{18.5}$ $= 768.64864865$ $= 770\ \text{km/h}$ (2 s f)	**1**	1 mark for correct answer (rounding not specified)

Question 35			
(a)	Share of rent $= \frac{492}{4} \times 2 = \$246.00$ Total annual utilities $= 4 \times (572 + 260) = 3328$ Fortnightly utilities for 4 residents $= \frac{3328}{26} = \$128$, so Sienna's share $= \frac{128}{4} = \$32$ Current fortnightly accom expenses $= 246 + 32 = \$278$ Fortnightly accom expenses increased by 20% $= \$278 \times 1.2 = \$333.60$	**2**	2 marks for correct answer 1 mark for correct total before increasing by 20% or equivalent merit
(b)	Fortnightly car loan payments $= 572 \times \frac{12}{26} = \$264.00$ Fortnightly running costs $= 48 \times 2 = \$96.00$ Fortnightly Serv/Ins/reg $= \frac{1831.70}{26} = \$70.45$ Total Fortnightly Car budget $= 264.00 + 96.00 + 70.45 = \$430.45$	**2**	2 marks for correct answer 1 mark for correct fortnightly amount for at least two of the component amounts or equivalent merit
(c)	Fortnightly food groceries $= 145 \times 2 = \$290$ Fortnightly Lunch and incidentals $= 40 \times 14 = \$560$ Total fortnightly budgeted $= 333.60 + 430.45 + 290.00 + 560.00$ $= \$1\,614.05$ Savings $= 1964 - 1614.05 = \$349.95$	**1**	1 mark for correct answer